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EXTINGO

Autonomous Fire Detection & Suppression Robot

System Logic & Flowchart Documentation

1. Overview

Extingo is an Arduino Uno-based autonomous fire-detection and suppression robot. On power-up it initializes its sensors and OLED display, then continuously scans its surroundings with a pan/tilt servo head fitted with an IR flame sensor. In parallel, it independently monitors temperature (via a DHT22 sensor) for a sudden rate-of-rise that can indicate fire even before flame is directly in view, and it refreshes an OLED status display every three seconds. When fire is confirmed — by IR detection combined with corroborating DHT22 or MQ2 smoke-sensor readings — the robot freezes its scan position, alerts the owner by SMS and phone call with GPS coordinates, activates smoke extraction, and suppresses the fire with a water pump (at reduced pressure if a person is detected nearby by the PIR sensor, or full pressure otherwise) while sounding an alarm. Once the IR sensor no longer detects flame, all suppression systems shut down and the robot returns to its scanning routine.

2. Hardware & Components

The table below lists every hardware component referenced in the flowchart, its role in the system, and how it connects to the Arduino Uno controller.

Component	Function in the System	Controlled / Read By
Arduino Uno	Main microcontroller running all logic and coordinating every sensor and actuator	Central controller
I2C OLED Display	Shows “Extingo Ready” at boot, then live sensor status every 3 seconds	Arduino (I2C bus)
IR Flame Sensor	Detects flame/fire signature while the scan head sweeps; primary fire trigger	Arduino (digital/analog input)
Servo Motor – Lower (Pan)	Sweeps the sensor head East–West during scanning; freezes on flame detection	Arduino (PWM)
Servo Motor – Upper (Tilt)	Tilts the sensor head North–South during scanning	Arduino (PWM)
DHT22 Sensor	Continuously samples temperature and humidity; feeds the rate-of-rise calculation	Arduino (digital input)
MQ2 Gas/Smoke Sensor	Detects smoke/combustible gas; one of three inputs to fire-confirmation logic	Arduino (analog input)
PIR Motion Sensor	Detects whether a person is present near the fire, to decide pump pressure	Arduino (digital input)
L298N Motor Driver	Drives the exhaust fan and water pump outputs (OUT1–OUT4)	Arduino (digital control pins)
12V Water Pump	Sprays water at LOW pressure (person present) or HIGH pressure (no person)	L298N OUT1/OUT2
Exhaust Fan	Extracts smoke/heat once a fast temperature rise or fire is confirmed	L298N OUT3/OUT4
GPS Module	Supplies geo-location coordinates attached to the fire alert message	Arduino (serial/UART)
GSM / SIM Module*	Sends SMS alerts and places the emergency call to the owner's number	Arduino (serial/UART)
ISD1820 Sound Module	Plays a looping siren/alarm sound once fire is confirmed	Arduino (trigger pin)

* The flowchart labels SMS/call actions but does not explicitly name the communication module; a GSM/SIM module (e.g. SIM800L) is the standard component used to provide this function and is listed here for completeness.

3. Diagram 1 — Startup & Flame-Scanning Flow

This flow covers everything from power-on to the moment a fire is confirmed. It is the robot's default, idle behaviour whenever no fire is present.

Extingo - Startup & Flame-Scanning Flow (Diagram 1 of 3)

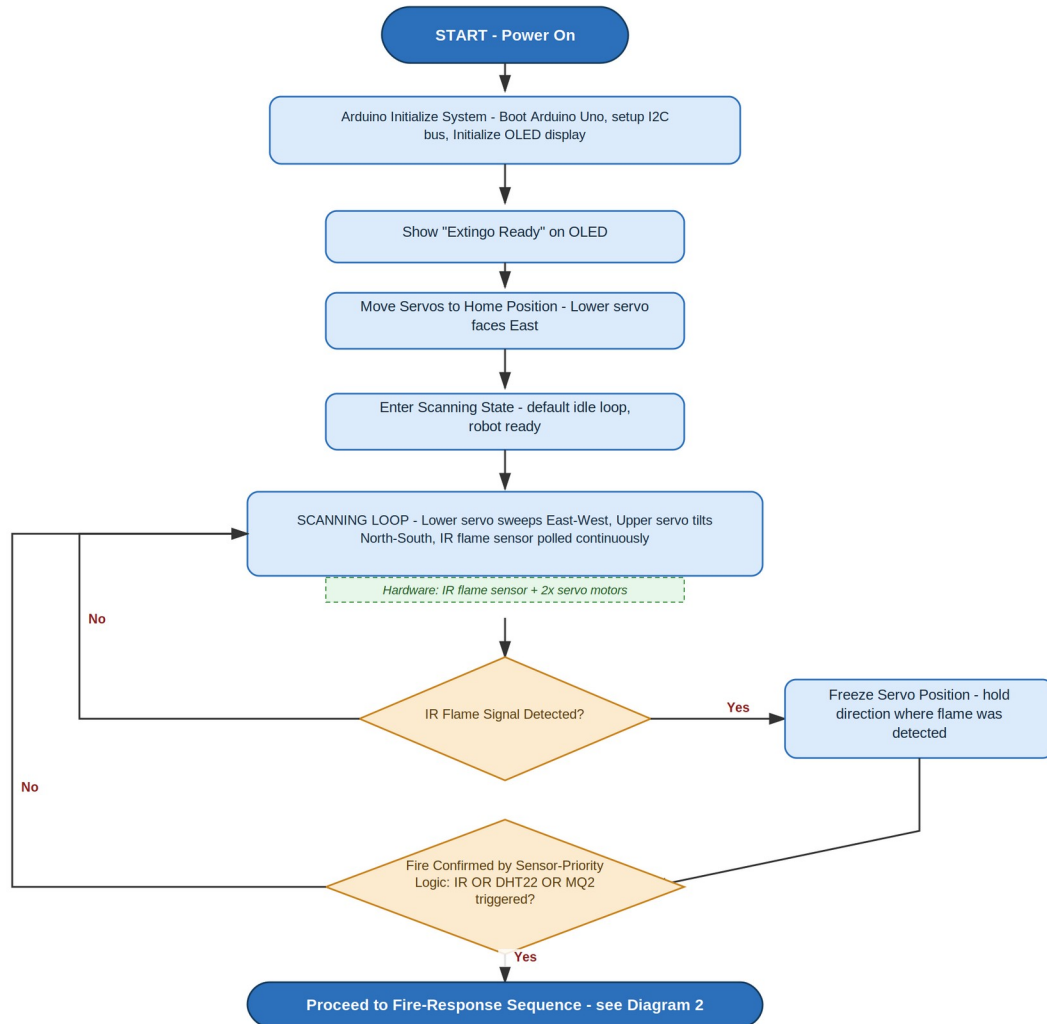


Diagram 1: Startup initialization, home positioning, and the continuous flame-scanning loop.

Step-by-step logic

1. Power on: the Arduino Uno boots.
2. System initialization: the Arduino sets up the I2C bus and initializes the OLED display.
3. The OLED shows “Extingo Ready” to confirm successful boot.
4. Both servos move to their home position, with the lower (pan) servo facing East.
5. The robot enters its Scanning State — the default idle loop while no fire is present.
6. Scanning loop: the lower servo sweeps East–West and the upper servo tilts North–South, while the IR flame sensor is polled continuously.
7. Decision — IR Flame Signal Detected? If No, the loop continues sweeping. If Yes, the servo position is frozen, holding the direction where the flame was detected.

8. Decision — Fire Confirmed by sensor-priority logic (IR OR DHT22 OR MQ2)? A single IR reading is not enough on its own; the system checks whether the DHT22 rate-of-rise or the MQ2 smoke sensor corroborates it. If No, the robot returns to scanning. If Yes, control passes to the Fire-Response Sequence (Diagram 2).

4. Diagram 2 — Fire-Confirmed Response Sequence

Once fire is confirmed, Extingo runs this sequence end-to-end: alerting the owner, extracting smoke, and suppressing the fire, before safely standing down and returning to scanning.

Extingo - Fire-Confirmed Response Sequence (Diagram 2 of 3)

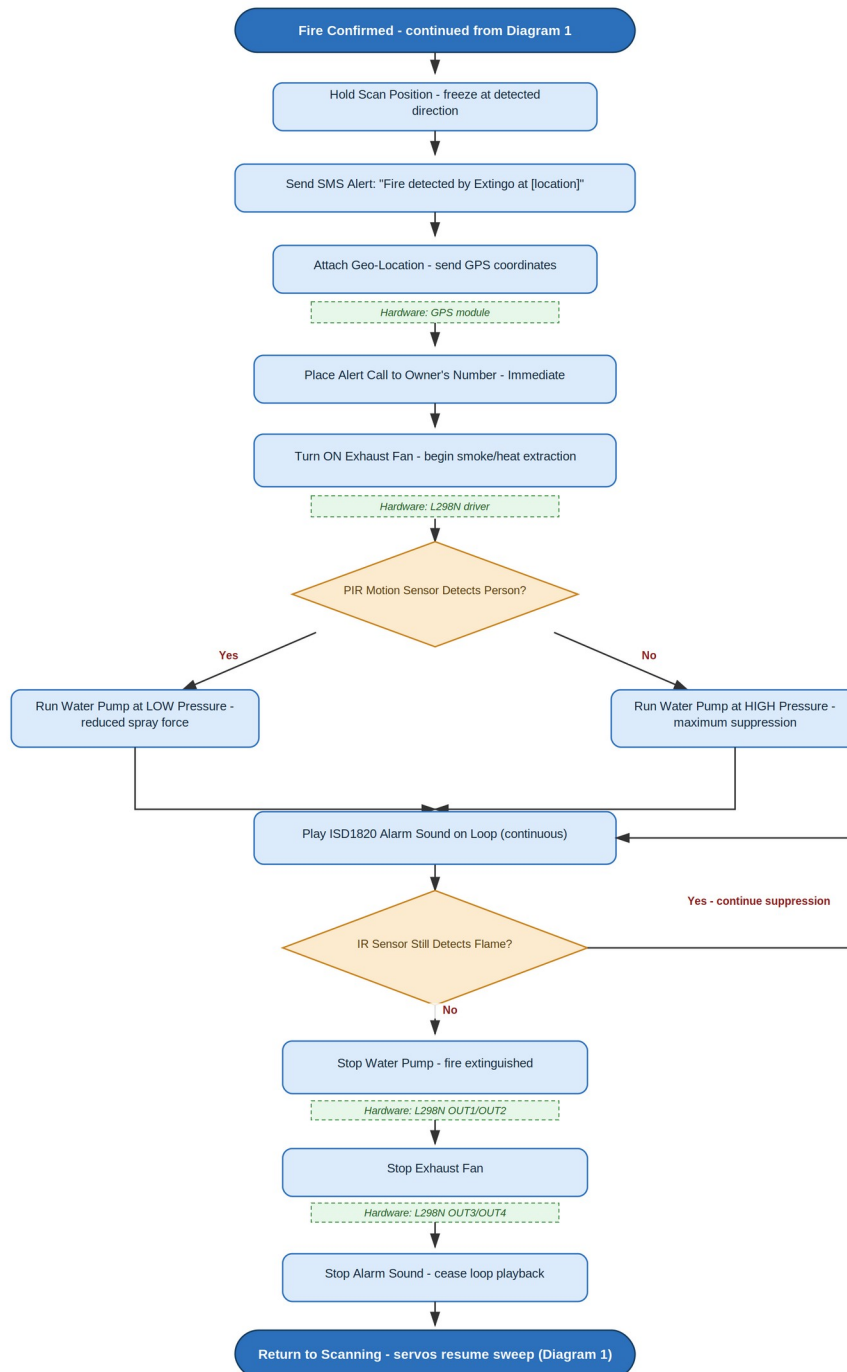


Diagram 2: Alerting, smoke extraction, and water-suppression sequence triggered on fire confirmation.

Step-by-step logic

1. Scan position is held/frozen at the direction where the flame was detected.
2. An SMS alert is sent: "Fire detected by Extingo at [location]."
3. Geo-location is attached using the GPS module, and coordinates are sent with the alert.
4. An emergency call is placed to the owner's number immediately.
5. The exhaust fan is turned on (via the L298N driver) to begin smoke and heat extraction.
6. Decision — does the PIR motion sensor detect a person nearby? If Yes, the water pump runs at LOW pressure to reduce spray force around a person. If No, the pump runs at HIGH pressure for maximum suppression.
7. The ISD1820 alarm sound plays on a continuous loop.
8. Decision — does the IR sensor still detect flame? If Yes, suppression continues (the pump keeps running and the alarm keeps looping). If No, the robot proceeds to shut down.
9. Shutdown sequence: the water pump stops (fire extinguished, via L298N OUT1/OUT2), the exhaust fan stops (via L298N OUT3/OUT4), and the alarm sound ceases.
10. The robot returns to Scanning, and the servos resume their sweep (back to Diagram 1).

5. Diagram 3 — Parallel Background Monitoring Loops

Two additional loops run continuously and independently of the main scanning/response flow: a DHT22-based rate-of-rise heat monitor, and an OLED status-display refresh loop.

Extingo - Parallel Background Monitoring Loops (Diagram 3 of 3)

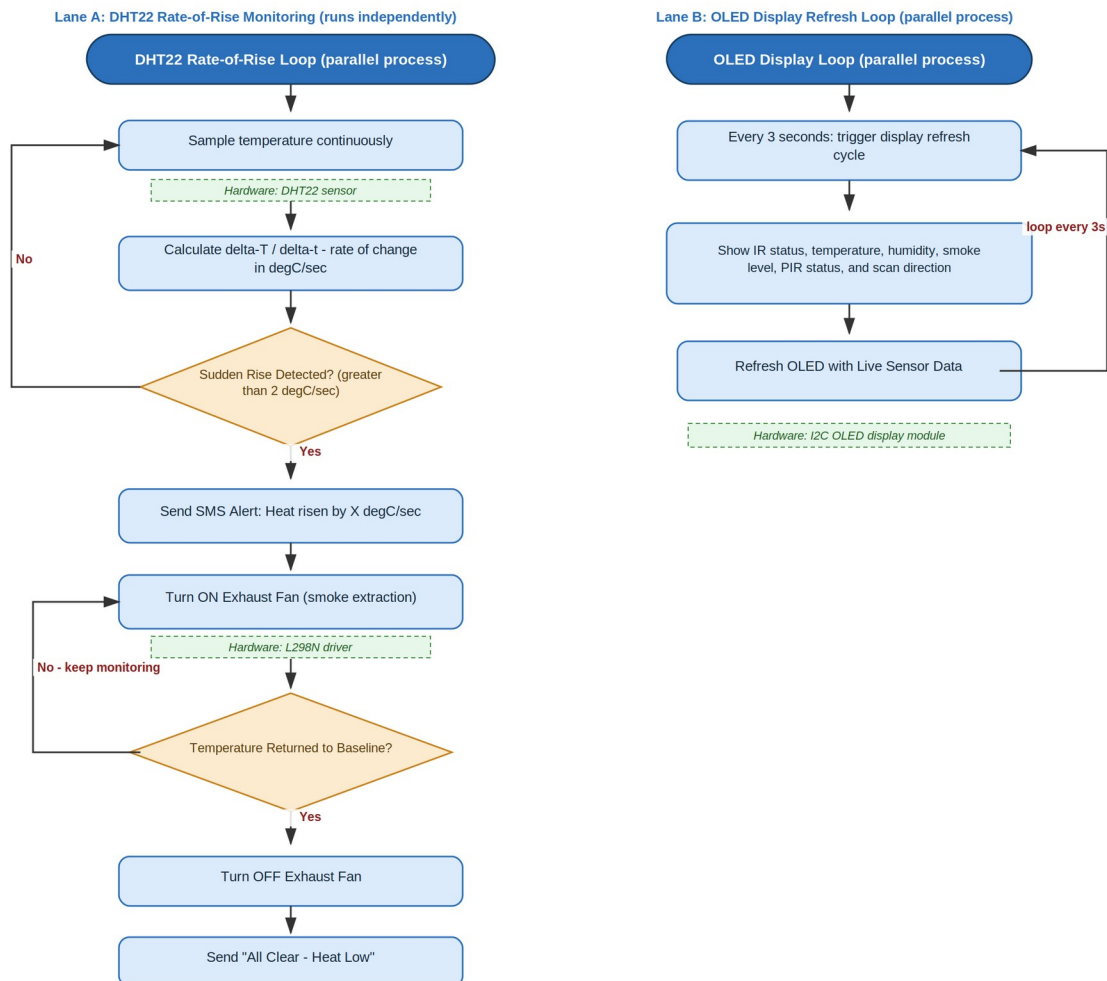


Diagram 3: The DHT22 rate-of-rise monitor (left) and the OLED display refresh loop (right) both run in parallel with scanning.

Lane A — DHT22 rate-of-rise monitoring

1. The DHT22 sensor continuously samples temperature (this loop runs independently of the scanning loop).
2. The system calculates $\Delta T/\Delta t$ — the rate of temperature change in °C per second.
3. Decision — is a sudden rise detected (greater than roughly 2°C/sec)? If No, sampling continues. If Yes, an SMS alert is sent (“Heat risen by X °C/sec”) and the exhaust fan is turned on via the L298N driver.
4. Decision — has the temperature returned to baseline? If No, the exhaust fan keeps running while monitoring continues. If Yes, the exhaust fan is turned off and an “All Clear – Heat Low” message is sent.

Lane B — OLED display refresh loop

1. Every 3 seconds, the display-refresh cycle is triggered.
2. The OLED shows live IR status, temperature, humidity, smoke level, PIR status, and current scan direction.
3. The display is refreshed with the latest live sensor data, then the loop repeats every 3 seconds.

6. Notes on This Reconstruction

The source file was an exported flowchart image (from the Boardmix diagramming tool) with overlapping and OCR-garbled text. The logic above has been reconstructed by matching each phrase to its most probable position in the flow, based on standard fire-detection-robot design patterns and the decision labels (Yes/No, “Parallel Process”) visible in the original. In particular:

- The exact SMS/location text and the 2°C/sec threshold are taken directly from legible fragments in the source; other numeric thresholds were not specified in the original and are not invented here.
- The GSM/SIM communication module is inferred, since SMS and calling functions require it even though the flowchart doesn't label it explicitly.
- If you have the original editable Boardmix project, re-exporting a higher-resolution or text-layer version would let this document be tightened up further — happy to revise it against that source.

thank you